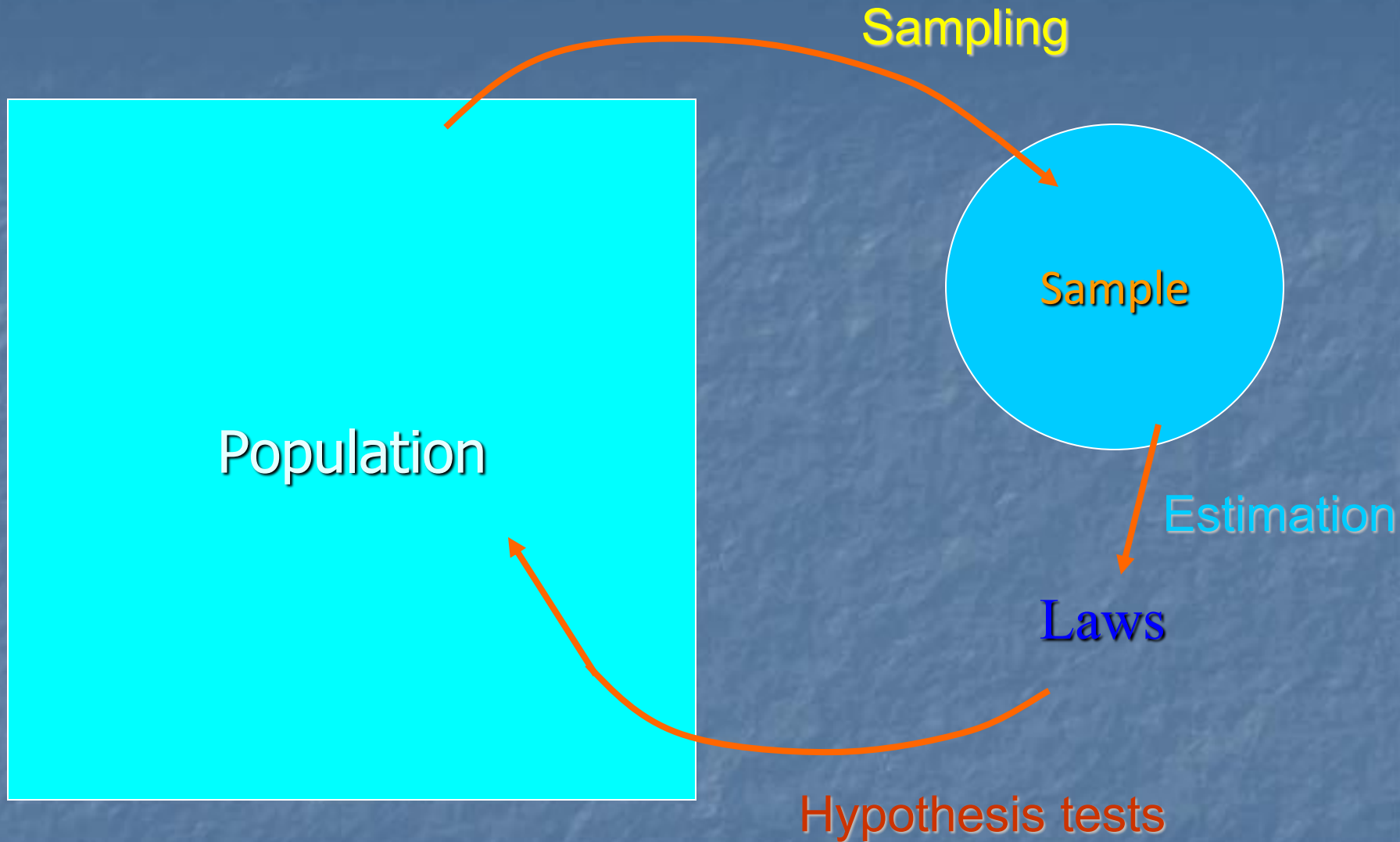


Parameter estimation

“**Estimation**”: Using **low accurate** measuring tools (using data collected in a **very limited sample** of population) to determine as precisely as possible value of a certain parameter (of all population).



Parameter Estimation

- * Estimation methods
- * Distribution of estimated parameters
- * Comparing distribution of estimated parameter with Normal distribution
- * Confidence Interval of estimation (Interval Estimation)

Estimation of rate (proportion, probability)

Example: - Tossing a coin: What is possibility to get “figure side”?

- Tossing a dice: What is probability to get the side with six points ?

- Tobacco smoking study: How large is smoking rate in elderly people (over 60) ?

- Proportion of rural households using rain water?

Usually it is very hard to determine exactly the real value of concerned parameter. The one must **estimate** the value by using some suitable method

→ Meet with some **error** in estimation

→ Need to evaluate **accuracy** of estimation: with a **given precise level** the estimation result is **acceptable** or not?

To determine **possible accuracy** of estimation with given precise level, we need to know **distribution** of the estimation

Distribution of variable

The set of values of a set of data, possibly grouped into classes, together with their frequencies or relative frequencies

Distribution of variable: the set of possible values with their probability

Example:

- **Tossing a coin:** Possibility to get "figure side" = $1/2$
→ uniform distribution of two values "figure side" and "number side"

- **Tossing a dice:** Probability to get the side with six points = $1/6$ → uniform distribution of 6 values

* , ** , *** , **** , ***** and *****

- **Tossing 6 dices:** Non-uniform distribution of 31 values
 6^* , 7^* , 8^* , ... , 35^* and 36^*

Concept of probability distribution

* **Discrete distributions:** *Variable X with*

Value:	X_1	X_2	X_3	$\dots$	X_n
Probability:	p_1	p_2	p_3		p_n

$$P \{X=X_1\} = p_1 \geq 0$$

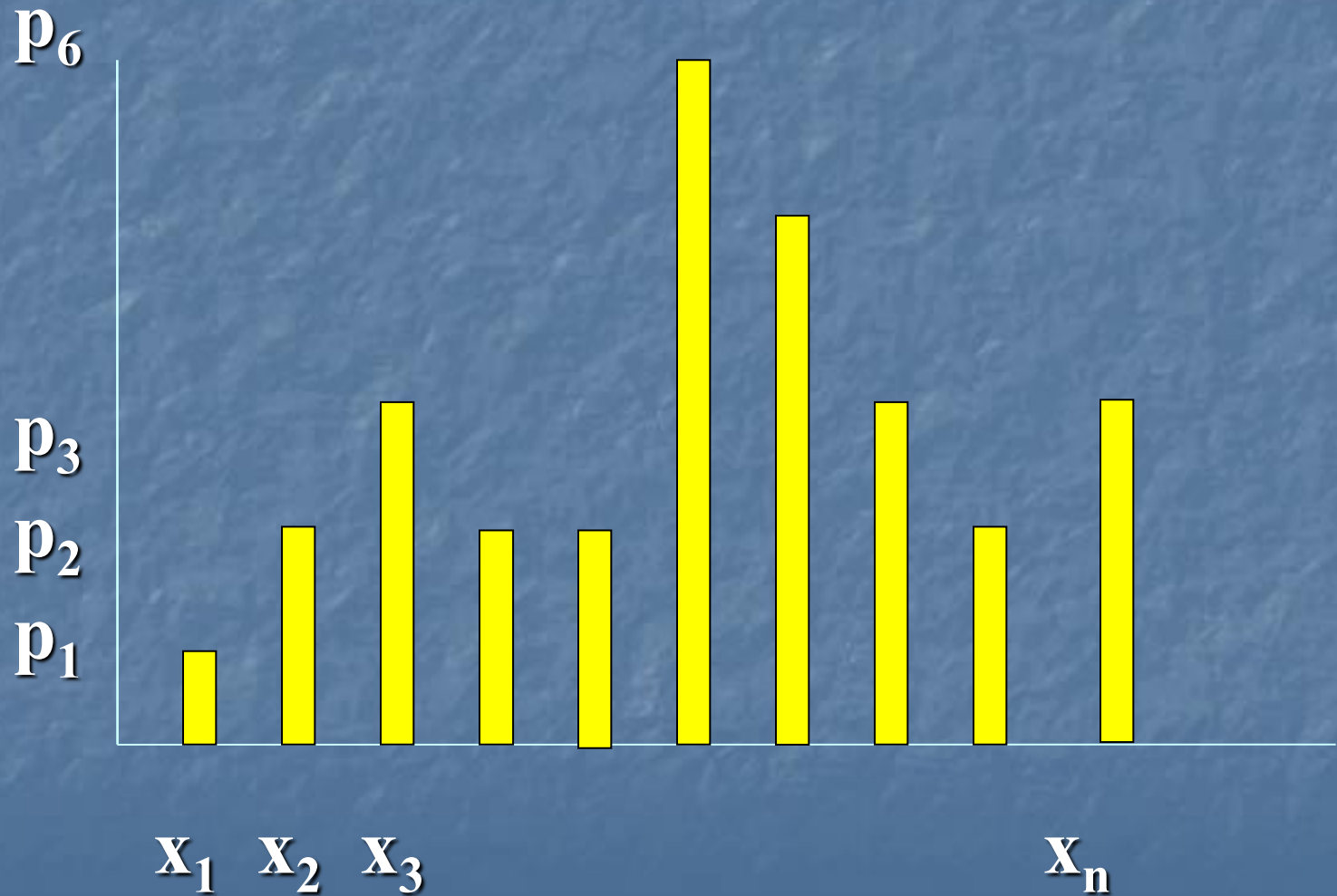
$$P \{X=X_2\} = p_2 \geq 0$$

$$\dots$$
$$P \{X=X_n\} = p_n \geq 0$$

$$p_1 + p_2 + \dots + p_n = 1 \text{ (100\%)}$$

Concept of probability distribution

* Discrete distributions:



Discrete Probability Distributions

The probability distribution is defined by a probability function, denoted by $f(x)$, that provides the probability for each value of the random variable.

Expected Value

- The expected value, or mean, of a random variable is a measure of its central location.

$$E(x) = \mu = \sum x f(x)$$

- The expected value is a weighted average of the values the random variable may assume. The weights are the probabilities.

Variance and Standard Deviation

- The variance summarizes the variability in the values of a random variable.

$$Var(x) = \sigma^2 = \sum (x - \mu)^2 f(x)$$

- The variance is a weighted average of the squared deviations of a random variable from its mean. The weights are the probabilities.
- The standard deviation σ is defined as the positive square root of the variance.

Concept of probability distribution

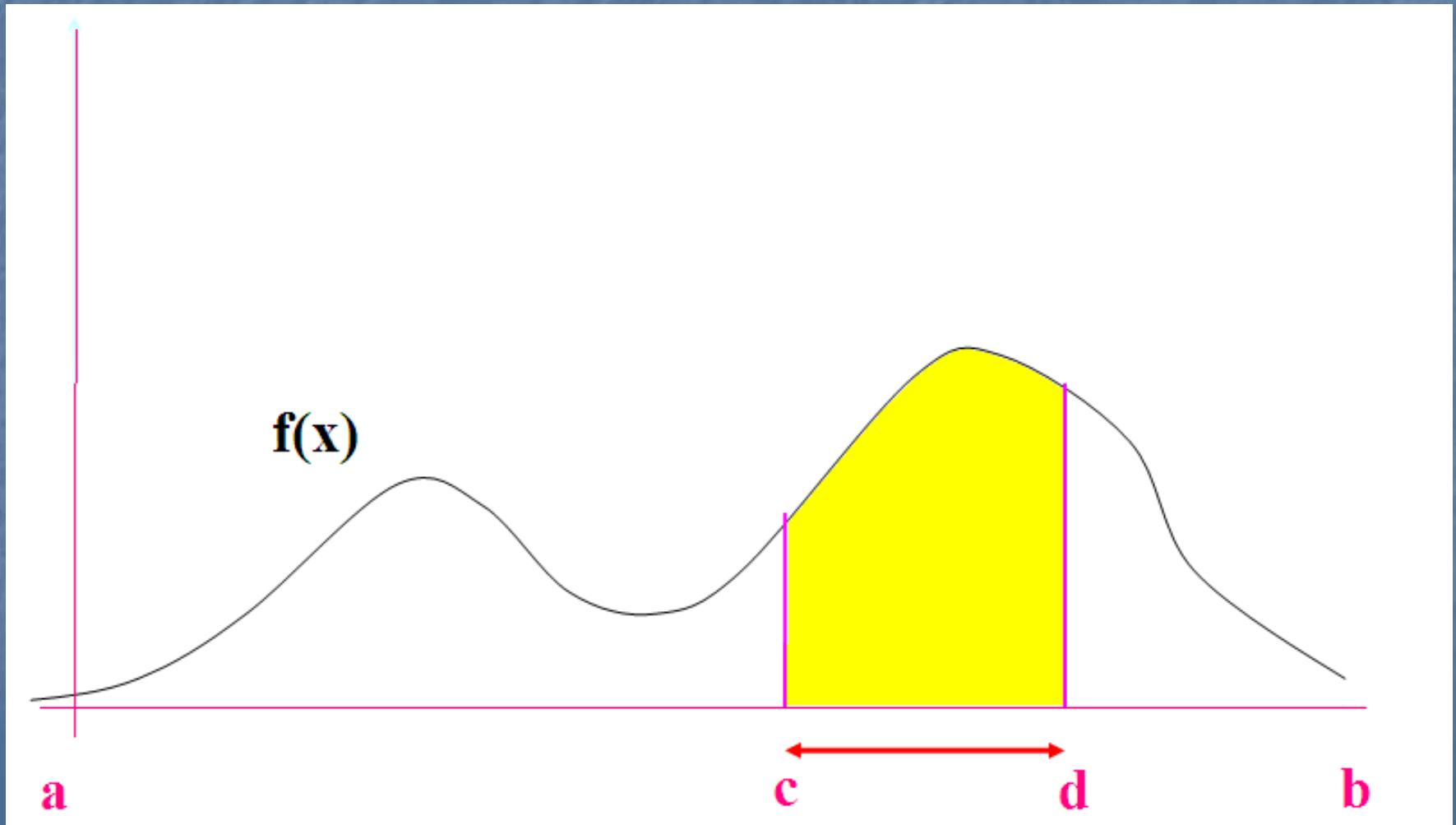
* **Continuous distributions:** *Variable X taken value x inside interval $(a;b)$ with density function $f(x) \geq 0$*

$$\int_a^b f(x)dx = 1 \quad ; \quad -\infty \leq a < b \leq +\infty$$

$$P\{X \in (c;d)\} = \int_c^d f(x)dx \quad \text{for } a \leq c < d \leq b$$

Concept of probability distribution

* Continuous distributions:



Expected Value

- The expected value, or mean, of a continuous random variable with density function $f(x)$:

$$E(X) = \mu = \int_a^b xf(x)dx$$

- The expected value is a “continuous weighted” average of the values the random variable.

Variance and Standard Deviation

- The variance summarizes the variability in the values of a random variable.

$$Var(X) = \sigma^2 = \int_a^b (x - \mu)^2 f(x) dx$$

- The variance is a continuously weighted average of the squared deviation of a random variable from its mean. The continuous weight is the probability density function.
- The standard deviation σ is defined as the positive square root of the variance.

Estimation of rate (proportion, probability)

In study population let's consider a binary variable X with two values 0 and 1

Suppose X takes value 1 with rate (proportion, probability) p and value 0 with rate $1 - p$, where p is unknown ($0 < p < 1$)

Estimation of rate (proportion, probability)

Usually we estimate the rate p by taking a sample of the variable X with n observations

$$x(1), x(2), \dots, x(n).$$

Determine the number $m(p)$ of values 1 among the n observations and perform the proportion

$$m(p) / n$$

as an estimated value of the rate p .

That way of estimation is "reasonable " or not?

THEOREM (**Louvl**ier). The proportion

$$m(p) / n$$

tends to **p** when **n** tends to infinity (is very large).

- The theorem proved mathematically shows the taking the proportion $m(p) / n$ for estimation of the rate **p** is completely “**reasonable**”: we can get the “**true**” rate when the **sample size** is very large.

Distribution of sample rate (proportion)

Let X be a binary variable taken value 1 with unknown probability p and taken value 0 with probability $1 - p$ (Bernoulli's distribution).

Estimating p : perform a sample $x(1), x(2), \dots, x(n)$ of X and take $m(p) / n$ as an estimation of p ($m(p)$ = number of 1's appeared in the sample).

Quantity $m(p) / n$ should take values $0/n, 1/n, 2/n, \dots, (n-1)/n, n/n$, each with certain “possibility” (probability)

Distribution of sample rate (proportion)

Quantity $m(p) / n$ is a random variable with **binomial distribution** with parameters p and n noted by $B(p;n)$

Binomial Distribution $B(p;n)$

$$P\{m(p) = k\} = C_n^k p^k (1-p)^{n-k}; \quad k = 0, 1, 2, \dots, n$$



Parameters of binomial distribution are the rate p and number n of experiments

Binomial Probability Distribution

- Expected Value

$$E(x) = \mu = np$$

- Variance

$$Var(x) = \sigma^2 = np(1 - p)$$

- Standard Deviation

$$SD(x) = \sigma = (np(1 - p))^{1/2}$$

Excel to Compute Binomial Probabilities

- BINOMDIST is used to compute the probability and cumulative probability.

Distribution of sample rate (proportion)

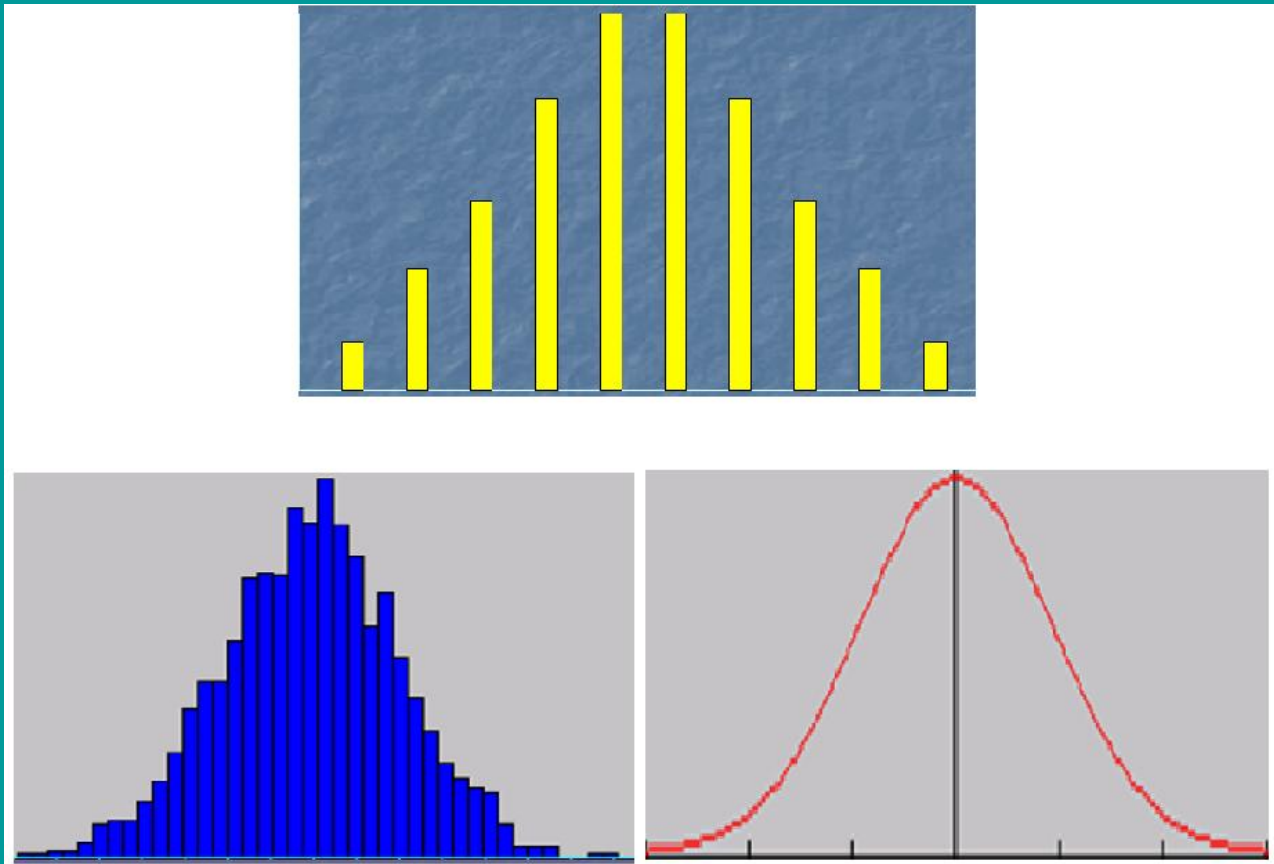
- Binomial distribution can be used to evaluate error in estimating p by $m(p) / n$
 - For small n , calculation with binomial distribution is practicable
 - For n large the calculation is very cumbersome → approximate Binomial distribution by Normal distribution

$$B(p;n) \sim N(p;p(1-p)/n)$$

Distribution of estimation of proportion

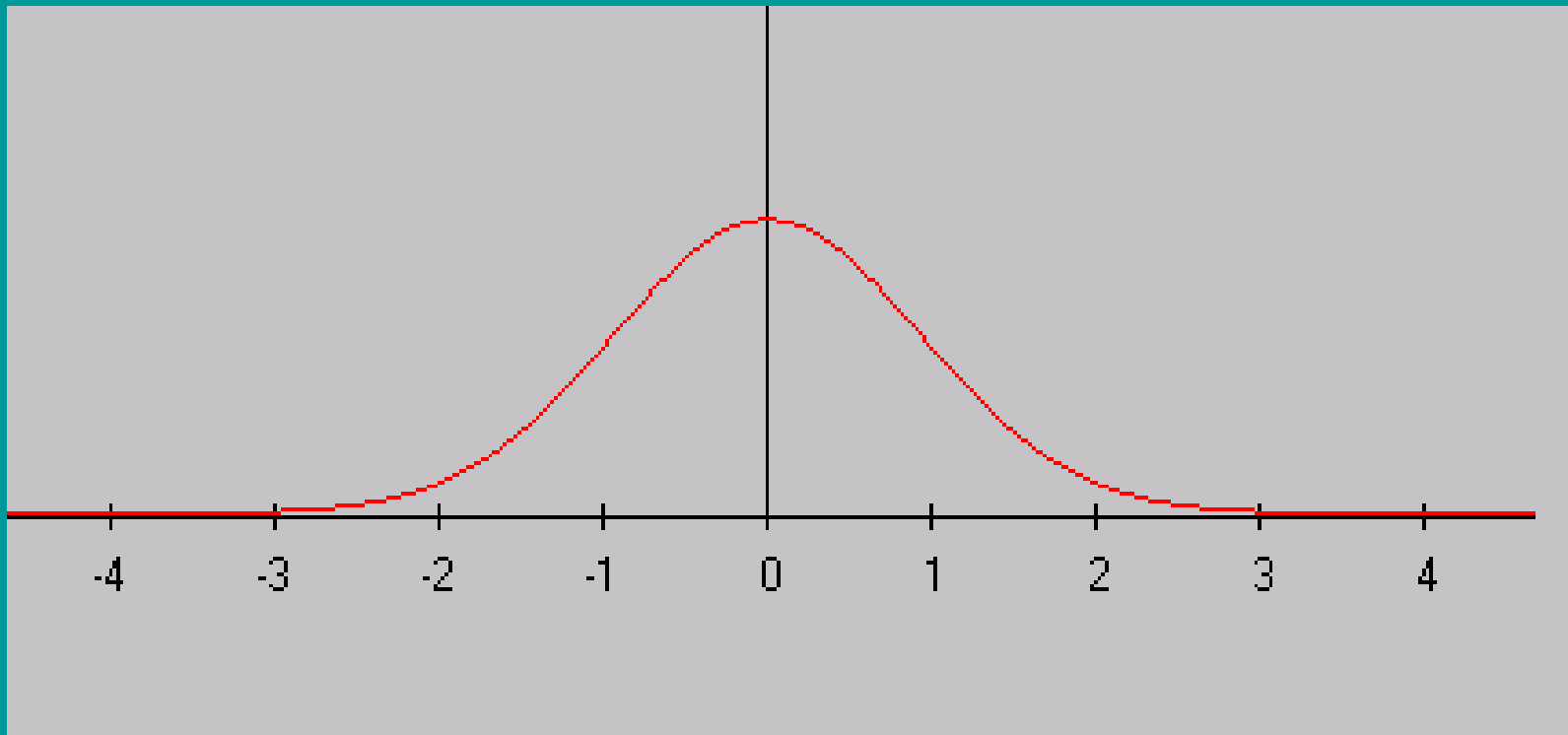
- The sampling distribution of $\bar{p}$ plays a key role in computing the margin of error for this interval estimate.
- The sampling distribution of $\bar{p}$ can be approximated by a normal distribution whenever $np \geq 5$ and $n(1 - p) \geq 5$.

Approximation of $B(p;n)$ by $N(p;p(1-p)/n)$



Normal distribution (Gaussian distribution) $N(\mu; \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{\sigma^2}\right)$$



Normal distribution defined by its expectation μ and variance σ^2

Normal distribution $N(\mu; \sigma^2)$

- Expected Value

$$E(X) = \mu$$

- Variance

$$Var(X) = \sigma^2$$

- Standard Deviation

$$SD(X) = \sigma$$

Standard normal distribution $N(0;1)$

- Expected Value

$$E(Z) = 0$$

- Variance

$$\text{Var}(Z) = 1$$

- Standard Deviation

$$\text{SD}(Z) = 1$$

Distribution transformation

- Linear transformation $Y = aX + b$

$$E(Y) = a.E(X) + b ;$$

$$\text{Var}(Y) = a^2\text{Var}(X) ; \text{SD}(Y) = |a|.\text{SD}(X)$$

- Standardization transformation

$$Y = (X - E(X)) / \text{SD}(X).$$

Then $E(Y) = 0 ; \text{Var}(Y) = 1 ; \text{SD}(Y) = 1$

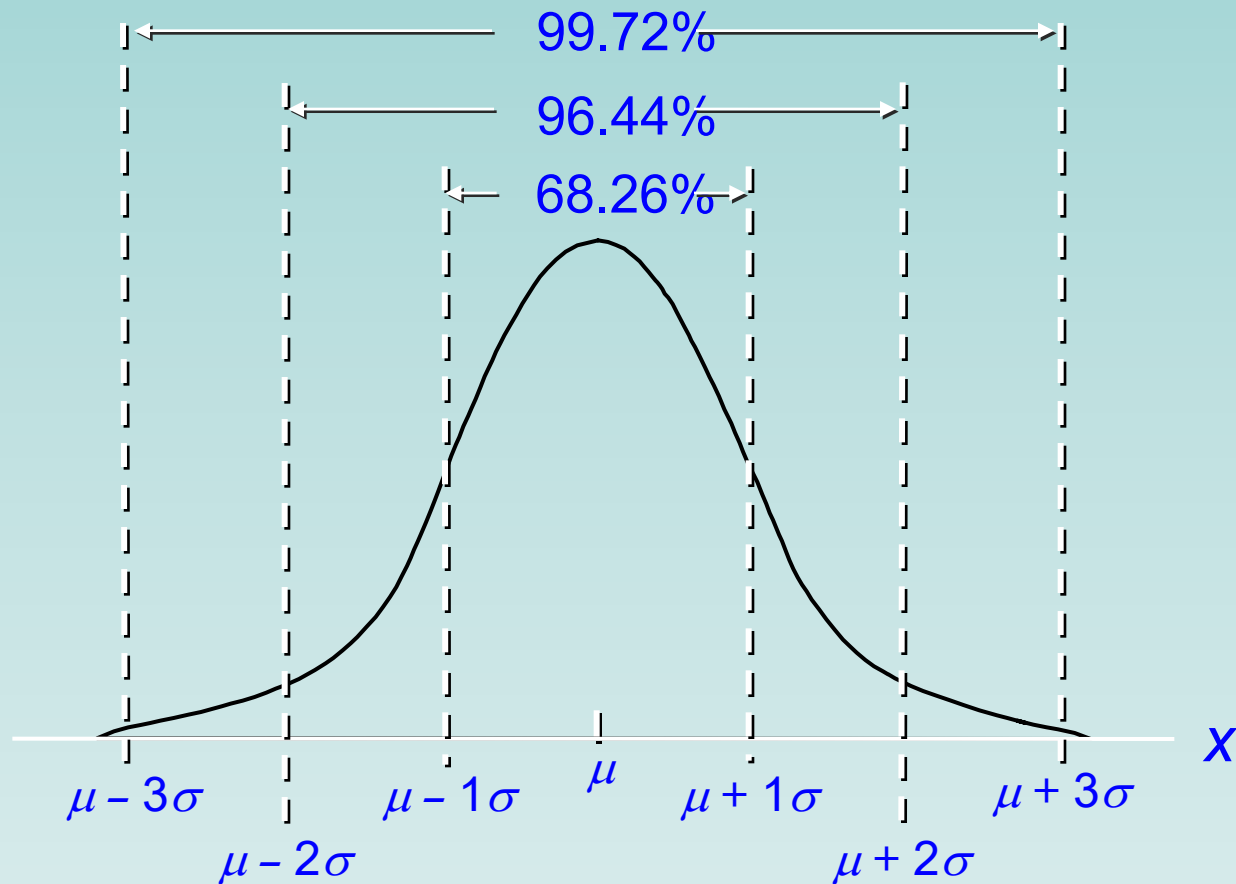
- Standardization of normal distribution $X \sim N(\mu; \sigma^2)$

$$Z = (X - \mu) / \sigma .$$

Then $E(Z) = 0 ; \text{Var}(Z) = 1 ; \text{SD}(Z) = 1$

Z is called standard normal, $Z \sim N(0;1)$

Normal Probability Distribution



Using Excel to Compute Normal Probabilities

- Excel has two functions for computing cumulative probabilities and x values for any normal distribution:
 - NORMDIST is used to compute the cumulative probability given an x value.
 - NORMINV is used to compute the x value given a cumulative probability.

- NORMDIST(x, MEAN,SD,cum) = Prob{X<x}
 - $X \sim N(\text{MEAN}, \text{SD}^2)$, cum = TRUE
- NORMSDIST(z, cum) = Prob{Z<z}
 - $Z \sim N(0,1)$, cum = TRUE

- NORMINV = inverse function of NORMDIST

NORMSINV = inverse function of
NORMSDIST

NORMSINV(cumulative-probability)

- $Z_{\alpha/2} = \text{NORMSINV}(1 - \alpha/2)$
- $\text{NORMSDIST}(Z_{\alpha/2}) = 1 - \alpha/2$

Confidence Interval of estimation

Confidence interval of an estimation is a segment on the number line that overlaps the estimated value of a parameter, indicating the true value of the parameter is likely to lie in that line with a given probability **a** (usually take **a = 95%**, $\rightarrow$ **95%CI**)



Confidence interval of estimation (interval estimation)

- **Confidence Interval** of estimation is an interval containing the estimated value of parameter, informing the **true value** of parameter can be some point inside the interval with given probability **a** .
- For a variable with normal distribution with expectation p and variance $p \cdot (1-p) / n$
95% Confidence Interval of estimation of p is the interval

$$(p - 1.96 * \sqrt{p \cdot (1-p) / n}; p + 1.96 * \sqrt{p \cdot (1-p) / n})$$

Confidence interval of proportion

Proportion is a quantity with distribution approximate to Normal Distribution,

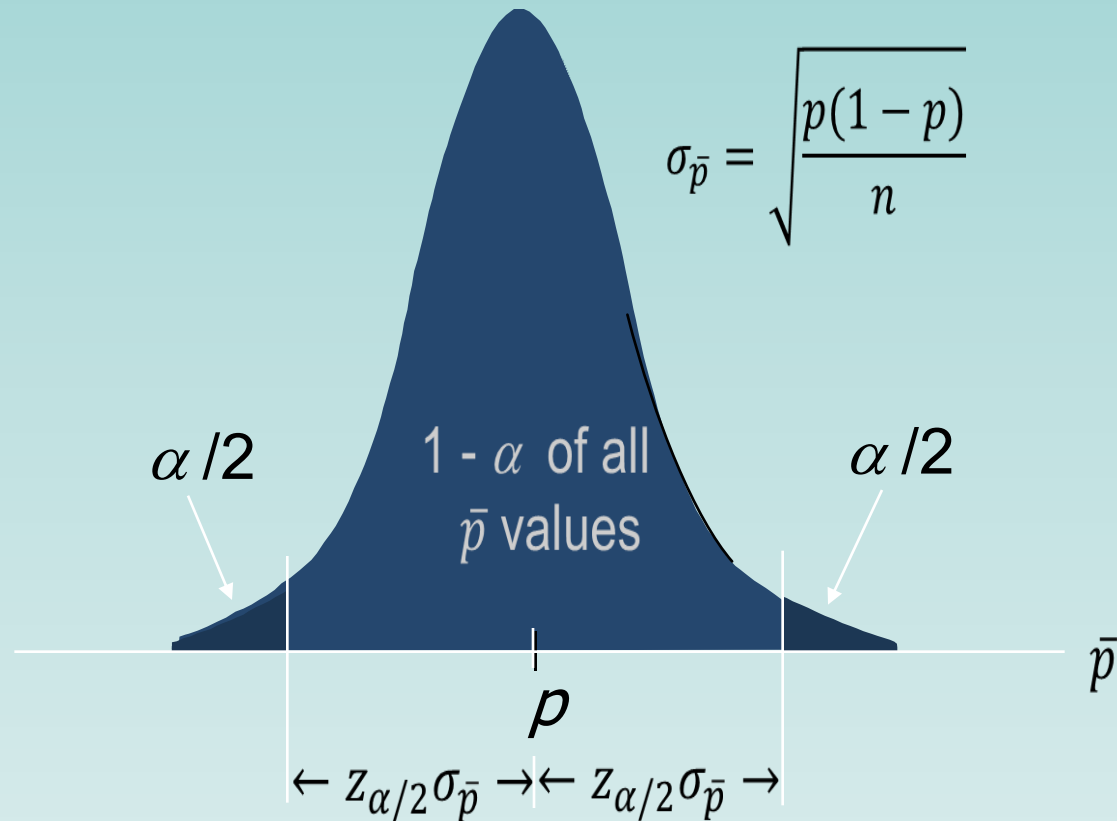
95% Confidence Interval of proportion is

$$\left[\hat{p} - 1.96 * \sqrt{\hat{p} \cdot (1 - \hat{p}) / n}; \hat{p} + 1.96 * \sqrt{\hat{p} \cdot (1 - \hat{p}) / n} \right]$$

where $\hat{p} = m(p) / n$

Interval Estimate of a Population Proportion

- Normal Approximation of Sampling Distribution of $\bar{p}$



Interval Estimate of a Population Proportion

- Interval Estimate

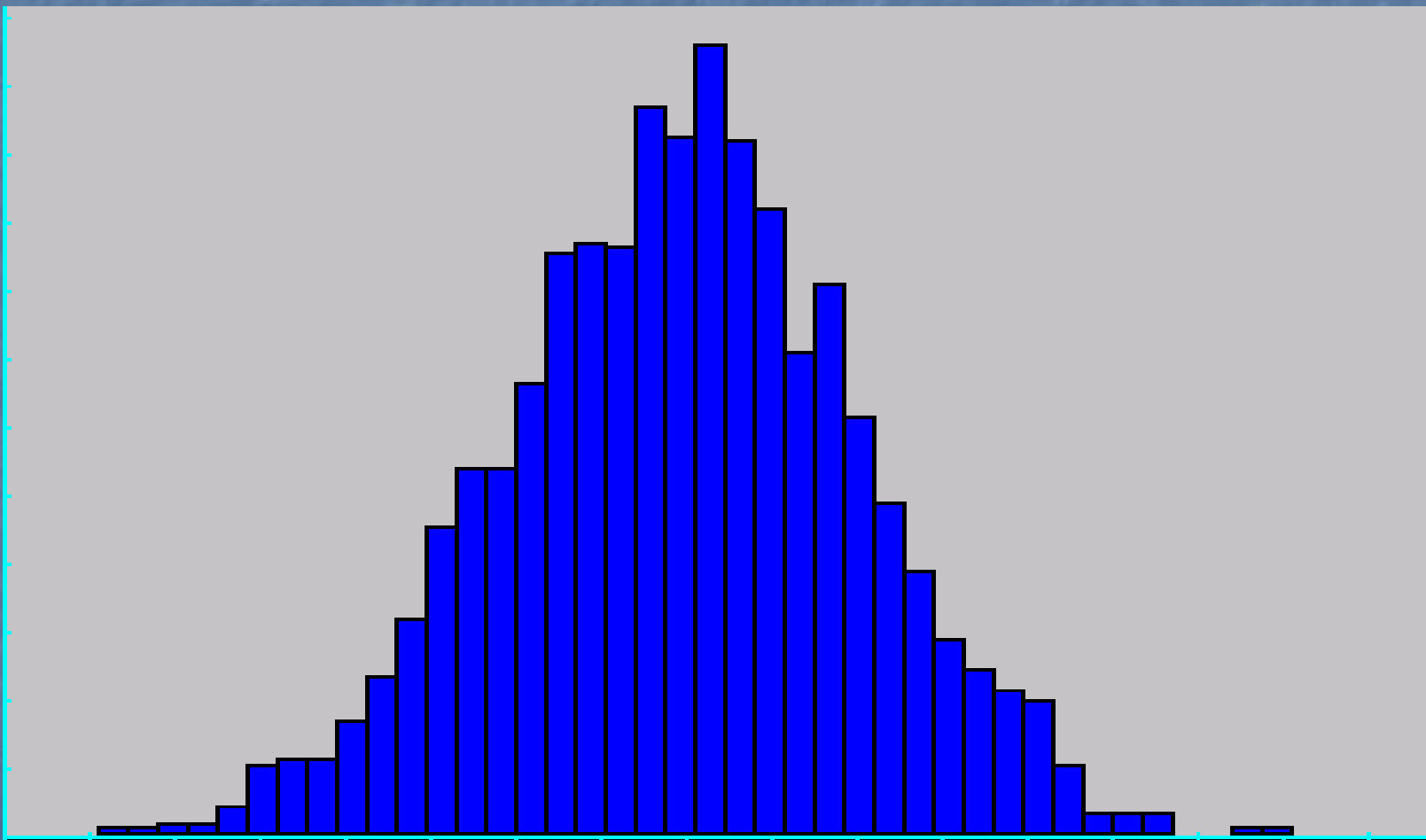
$$\bar{p} \pm z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

where: $1 - \alpha$ is the confidence coefficient,

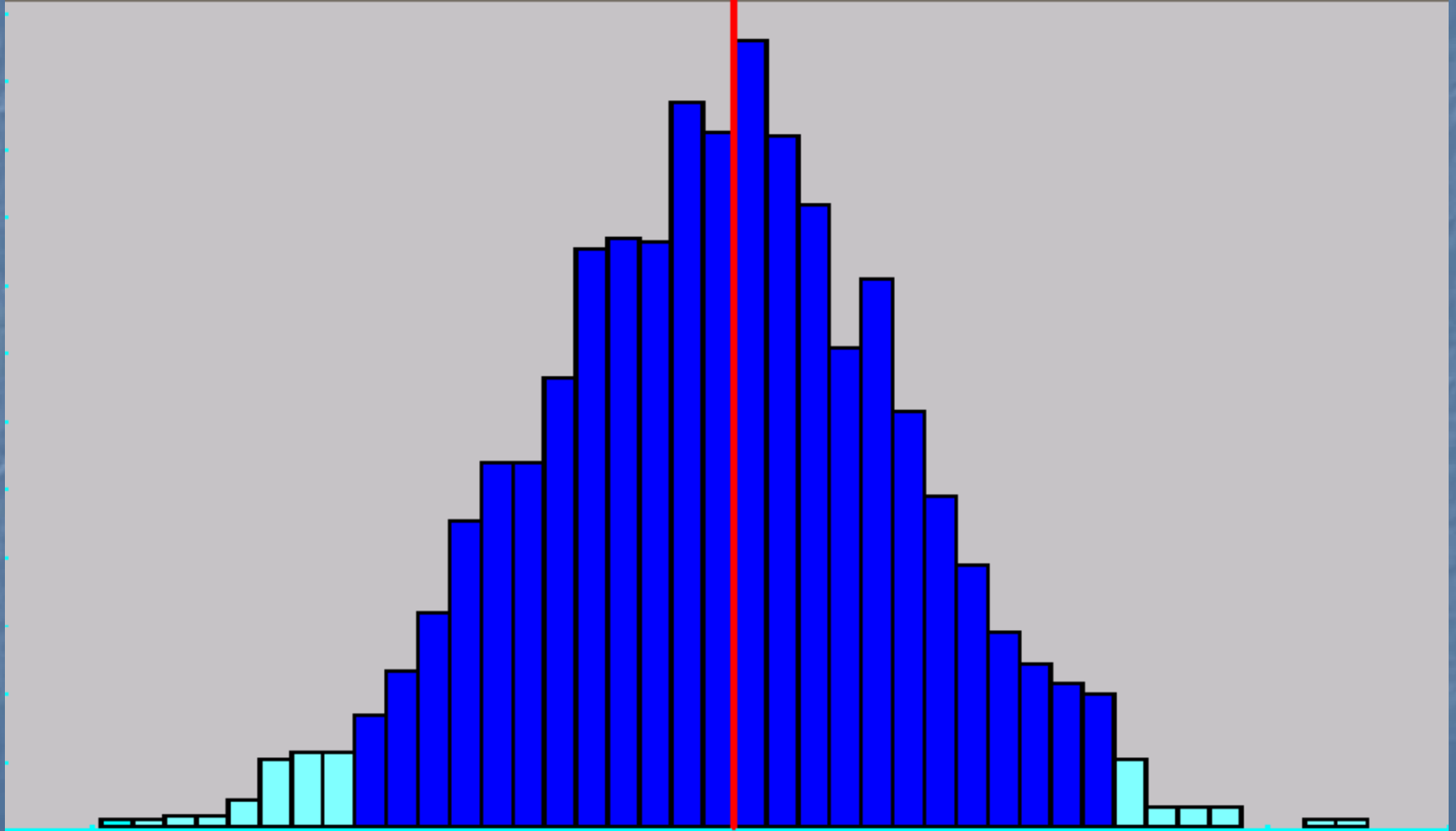
$z_{\alpha/2}$ is the z value providing an area of $\alpha/2$
in the upper tail of the standard normal
probability distribution, and

$\bar{p}$ is the sample proportion

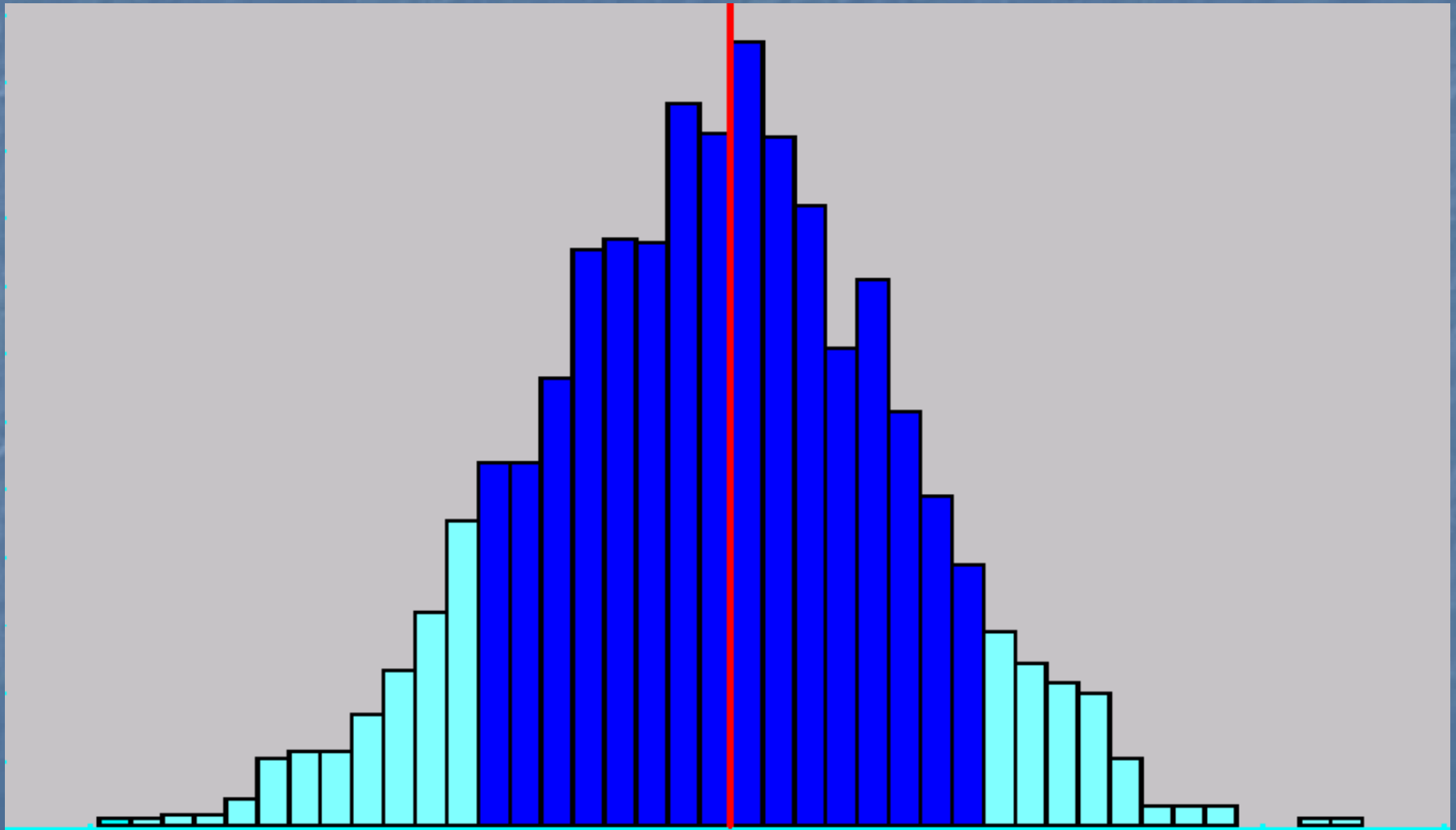
Normal distribution



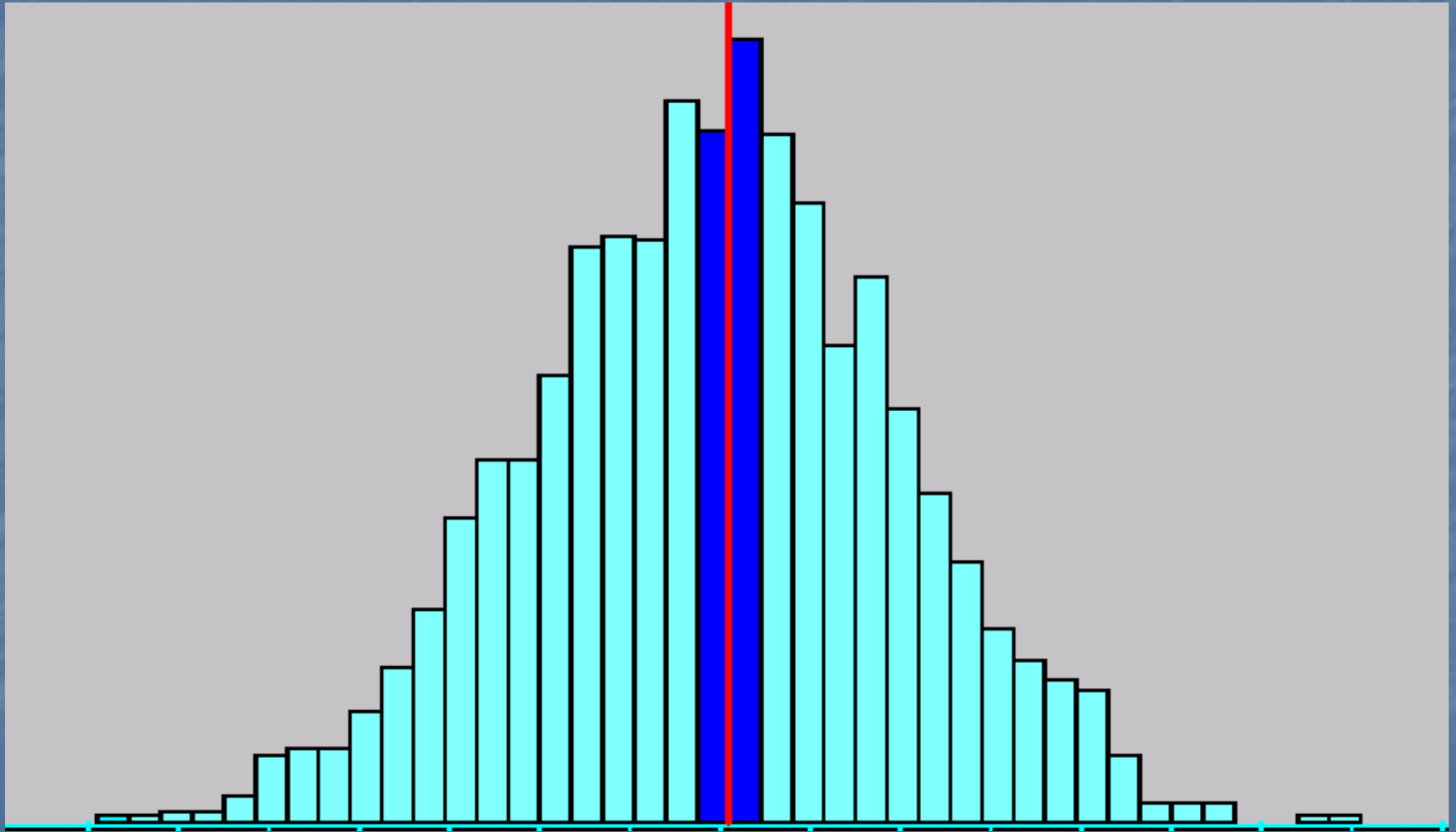
95% Confidence Interval



70% Confidence Interval



10% Confidence Interval



EXCEL PRACTICE

- $z_{\alpha/2} = \text{NORMSINV}(1 - \alpha/2)$
- $z_{\alpha/2} = \text{NORM.S.INV}(1 - \alpha/2)$
-

Estimation of Expectation

Expectation of variable =
Mean value of variable in whole population

To estimate the expectation of a quantitative variable X , a sample $x(1), x(2), \dots, x(n)$ is selected and **sample mean value** (sample average)

$$\bar{X} = \frac{1}{n} (x(1) + x(2) + \dots + x(n))$$

can be taken as an estimated value of **expectation parameter** $E(X)$ of X

Conclusion : Sample mean value is a “good” estimation of Expectation:

$$Mean_n(X) \xrightarrow{n \rightarrow \infty} E(X)$$

The estimation is very close to true value of expectation if sample size ***n*** is very large.

Problem: Although **Sample mean value** is a “good” estimation of **Expectation**, there exists always some error of that estimation

→ How to evaluate the error in that estimation?

→ Need to know about **distribution** of sample mean value

Distribution of sample mean value

THEOREM. Let variable X have normal distribution with expectation μ and variance σ^2 and select a sample $x(1), x(2), \dots, x(n)$ of that variable. Then the sample mean value

$$\bar{X} = \frac{1}{n} (x(1) + x(2) + \dots + x(n))$$

is a quantity with normal distribution with mean value equal μ and variance equal σ^2 / n .

Interval Estimate of a Population Mean:

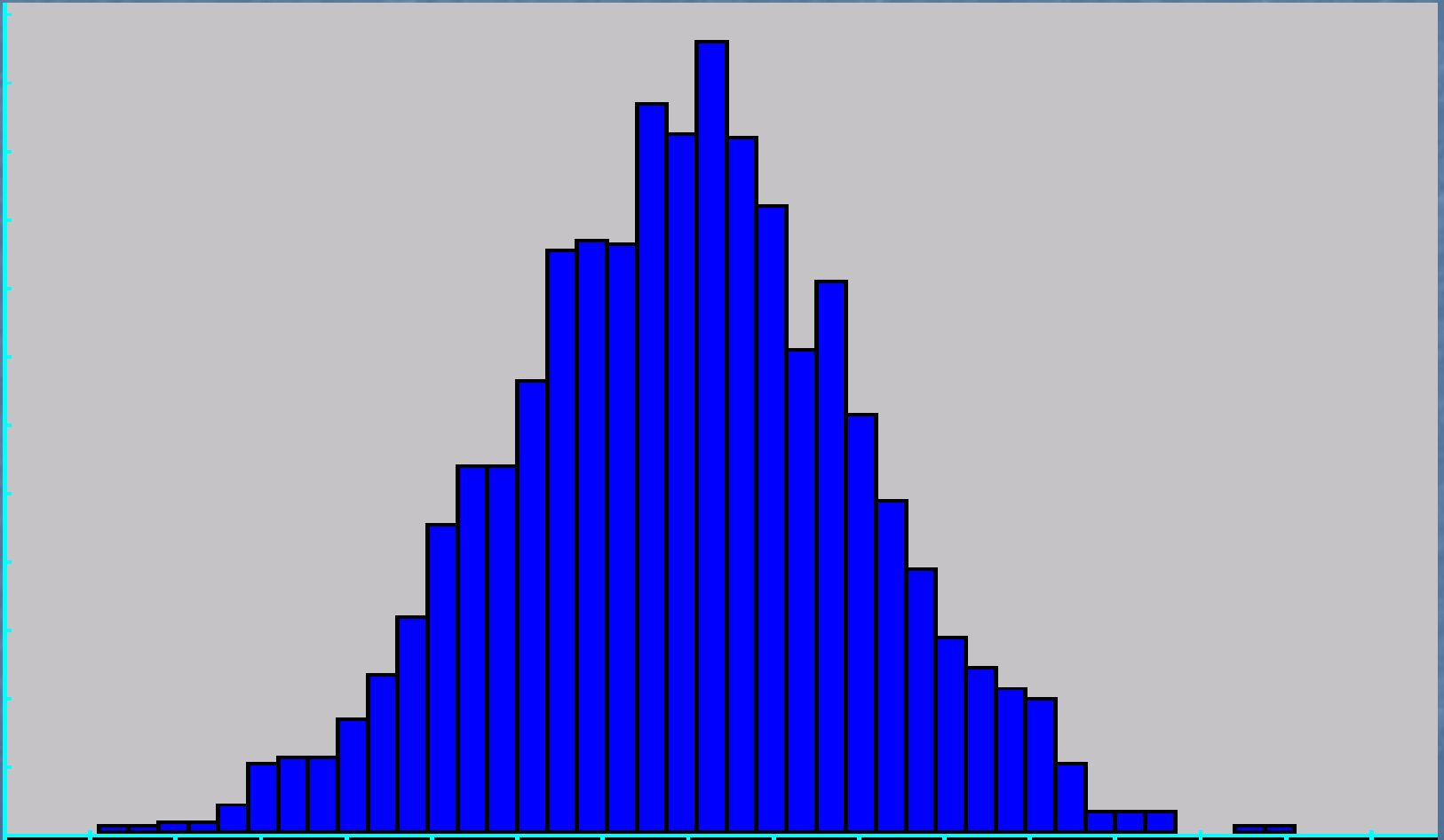
- In order to develop an interval estimate of a population mean, the margin of error must be computed using either:
 - the population standard deviation σ , or
 - the sample standard deviation s
- σ is rarely known exactly, but often a good estimate can be obtained based on historical data or other information.
- We refer to such cases as the σ known case.

Confidence Interval of sample mean value

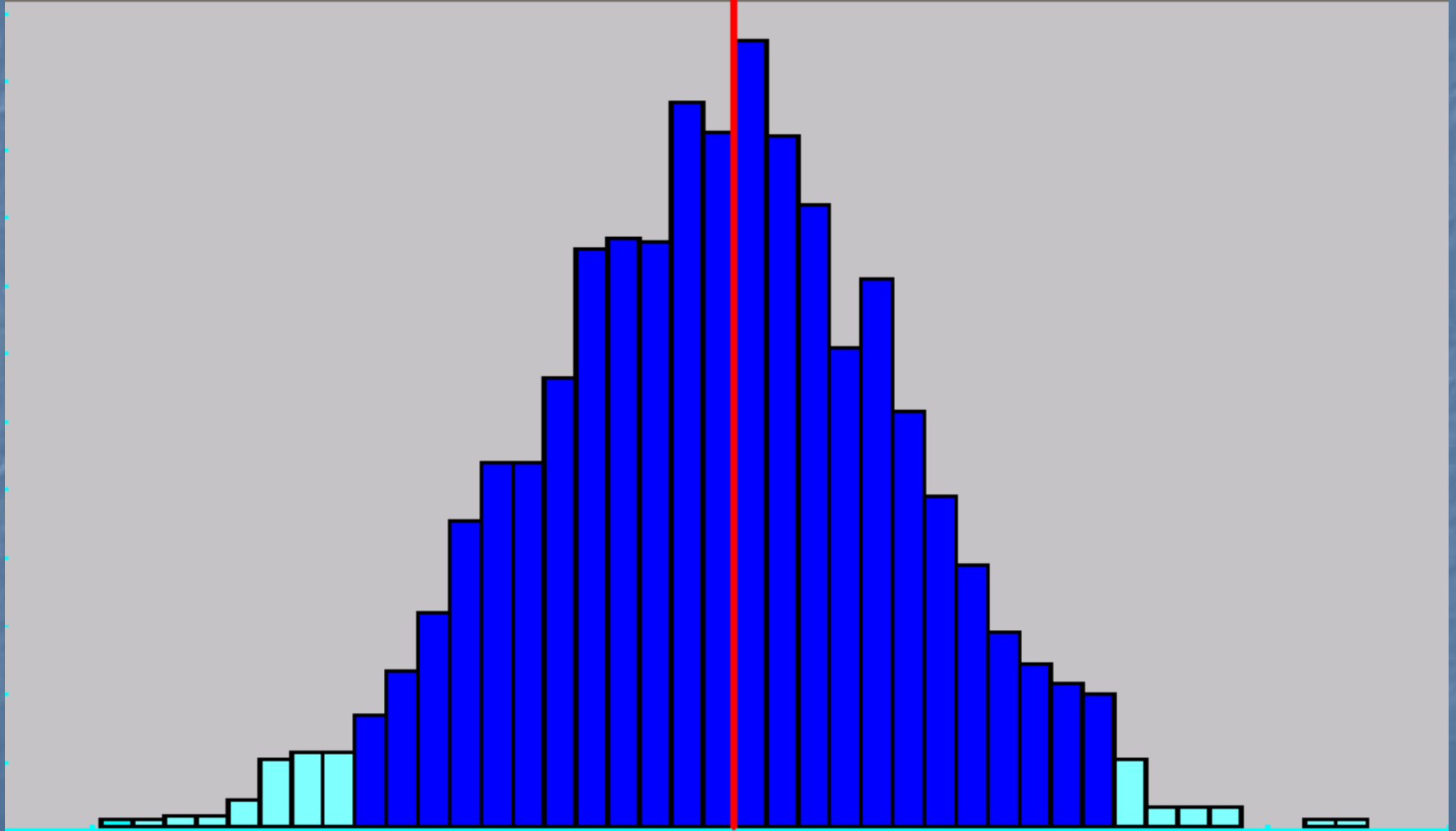
- **Confidence Interval** of estimation is an interval containing the estimated value, confirming the true value of estimated parameter should be a point of that interval with a given probability **a**
- For a normal distributed estimation quantity with expectation $\bar{X}$ and variance σ^2 / n , the **95% Confidence Interval** ($a = 95\%$) is defined by

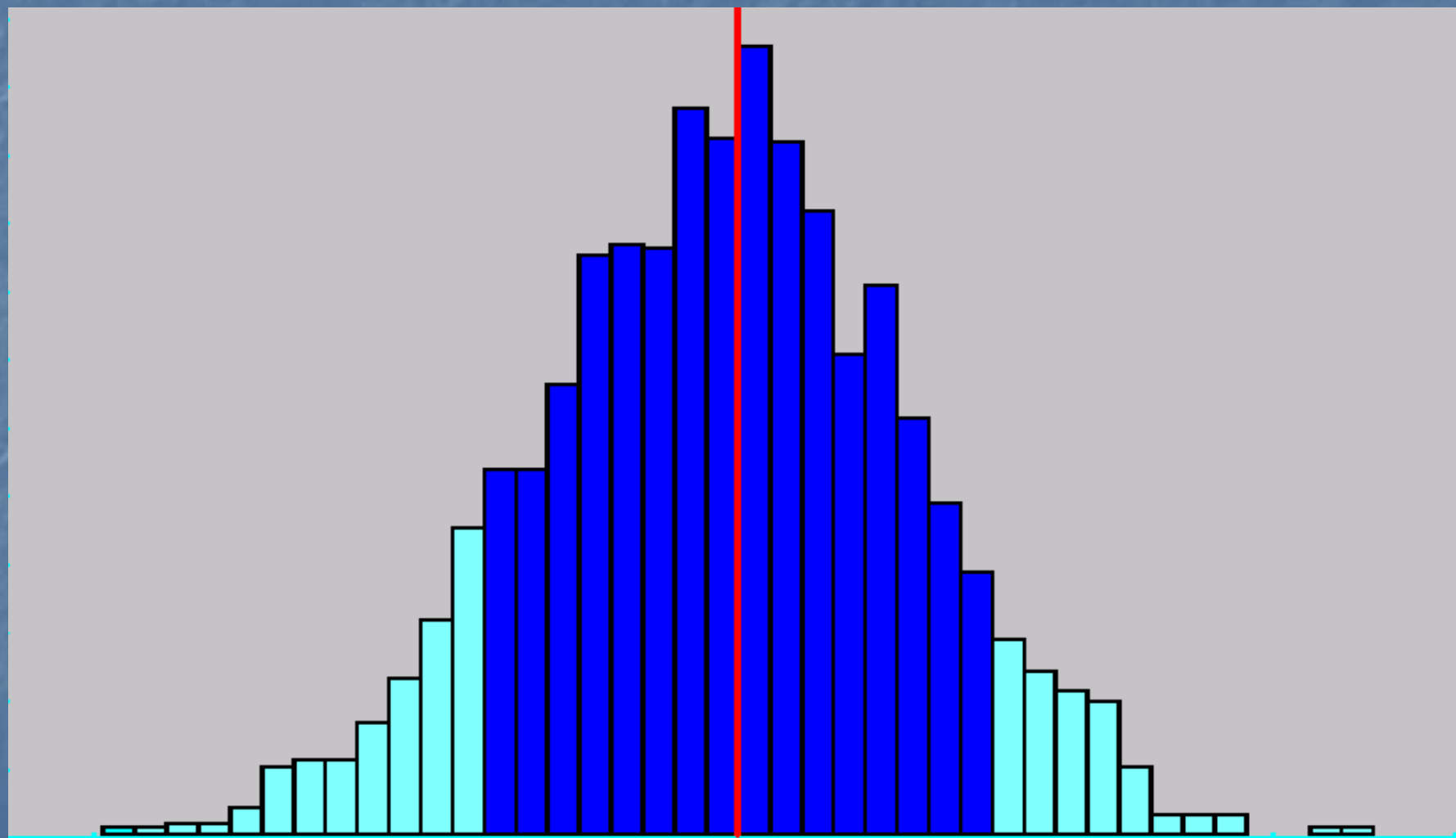
$$\left[\bar{X} - 1.96 * \sigma / \sqrt{n}; \bar{X} + 1.96 * \sigma / \sqrt{n} \right]$$

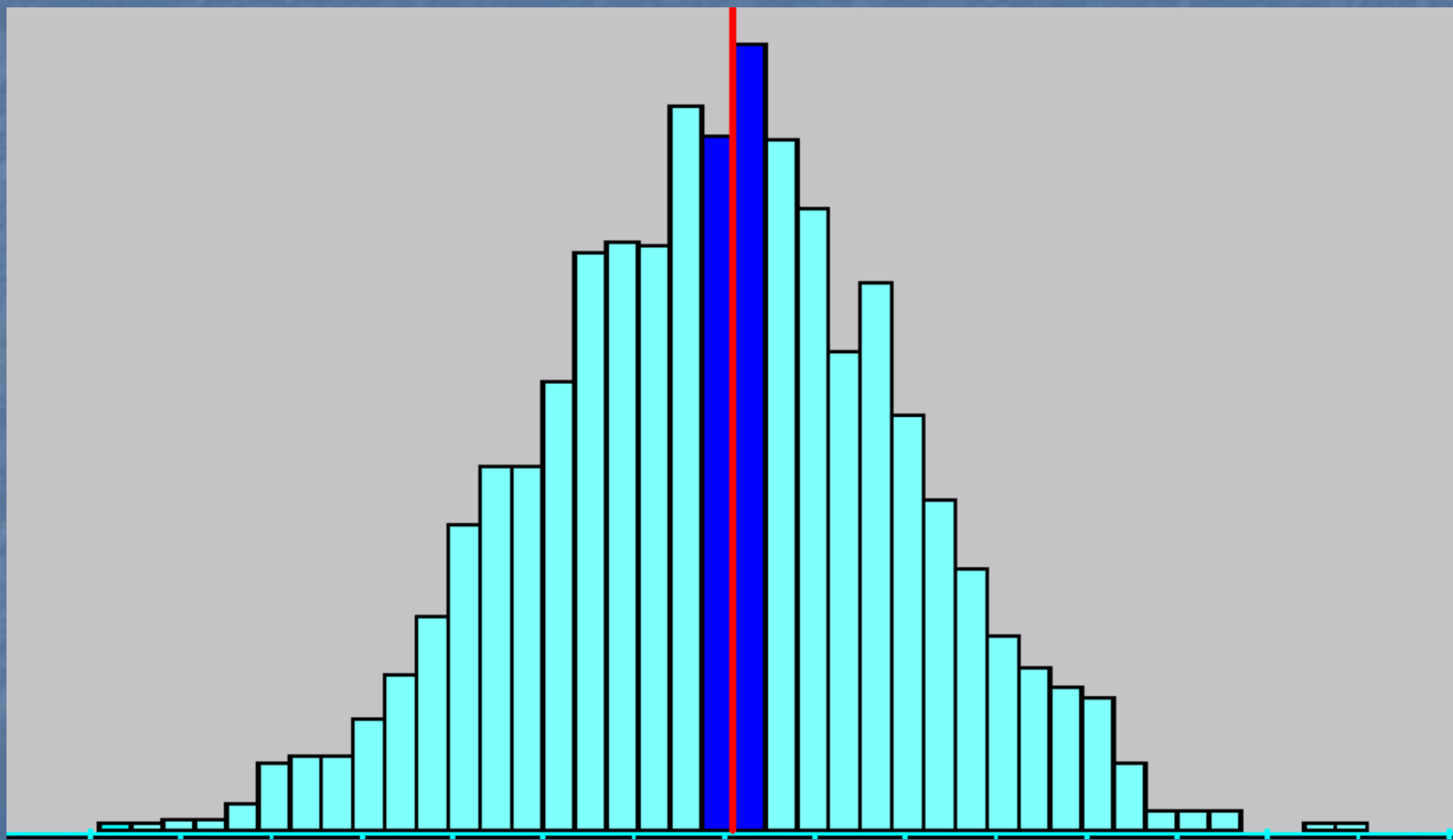
Normal distribution



Confidence Interval

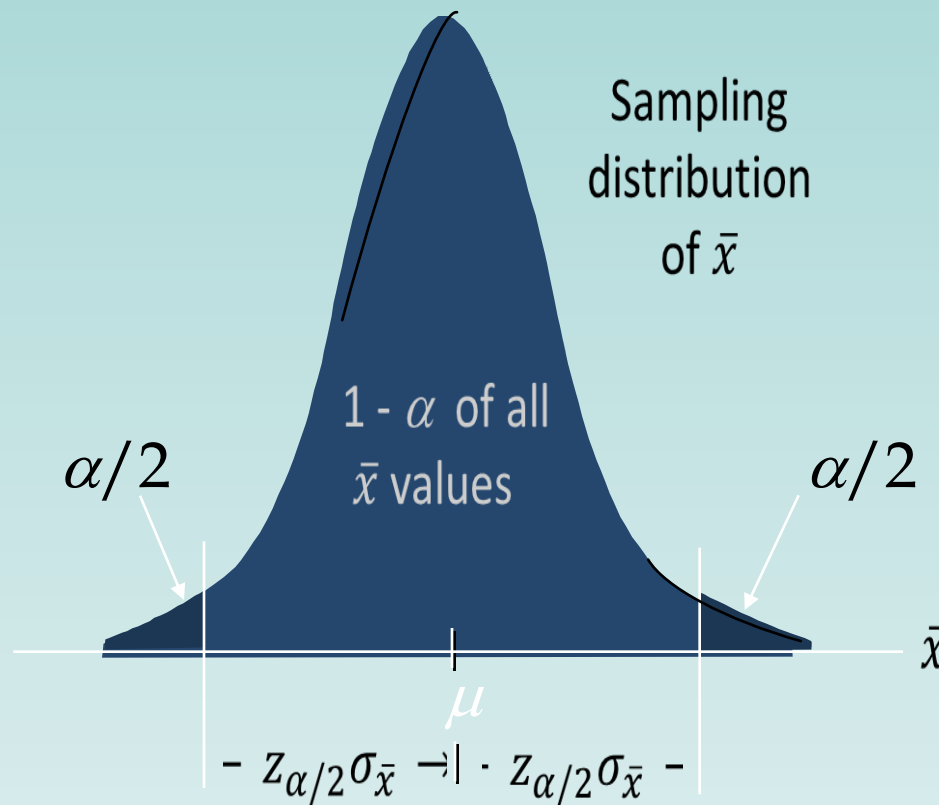




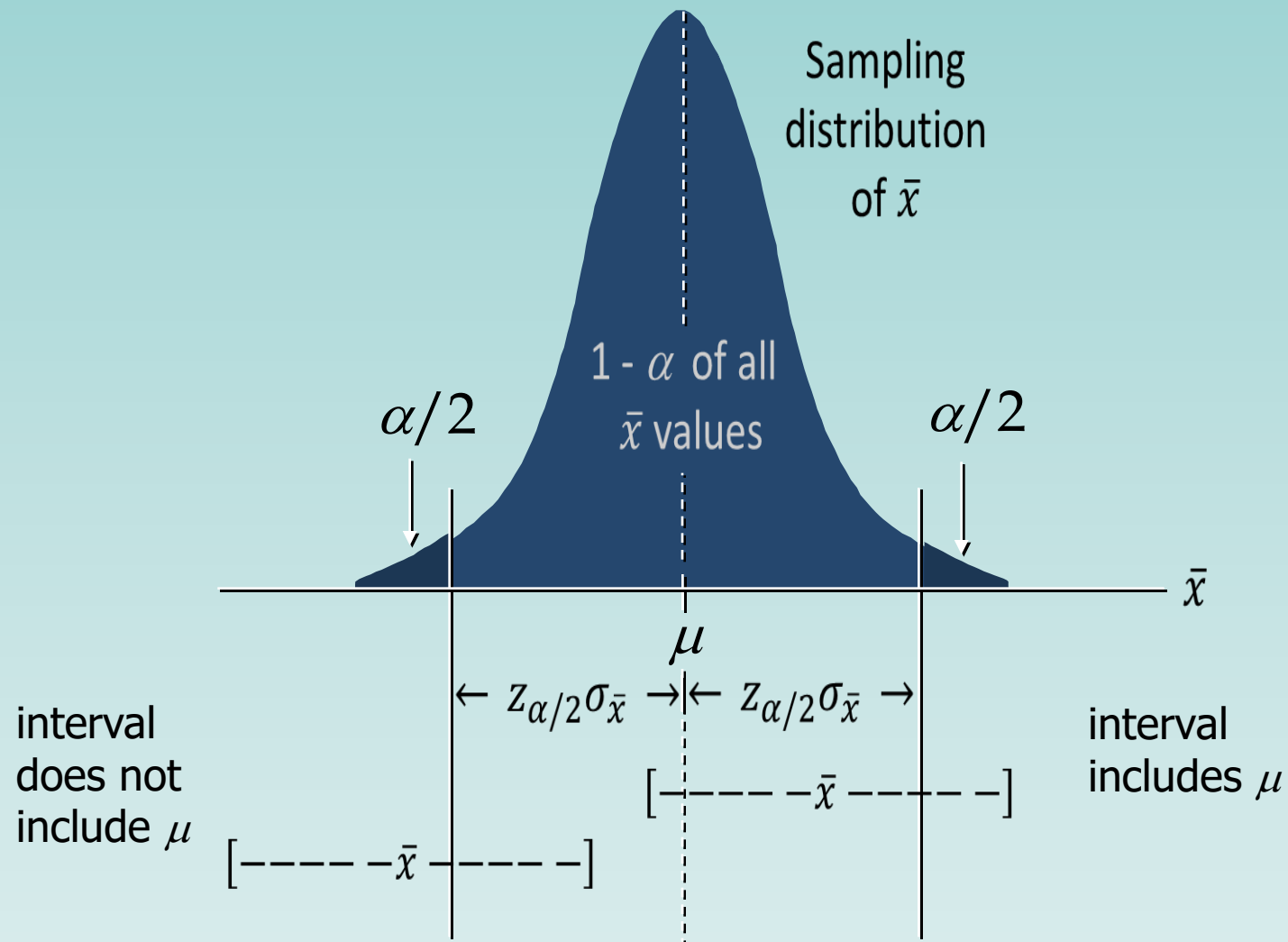


Interval Estimate of a Population Mean: σ Known

There is a $1 - \alpha$ probability that the value of a sample mean will provide a margin of error of $z_{\alpha/2}\sigma_{\bar{x}}$ or less.



Interval Estimate of a Population Mean: σ Known



Interval Estimate of a Population Mean: σ Known

- Interval Estimate of μ

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where: $\bar{x}$ is the sample mean

$1 - \alpha$ is the confidence coefficient

$z_{\alpha/2}$ is the z value providing an area of $\alpha/2$ in the upper tail of the standard normal probability distribution

σ is the population standard deviation

n is the sample size

Interval Estimate of a Population Mean: σ Known

- Values of $z_{\alpha/2}$ for the Most Commonly Used Confidence Levels

Confidence Level	α	$\alpha/2$	Table Look-up Area	$z_{\alpha/2}$
90%	0.10	0.05	0.9500	1.645
95%	0.05	0.025	0.9750	1.960
99%	0.01	0.005	0.9950	2.576

Meaning of Confidence

- Because 90% of all the intervals constructed using $\bar{x} \pm 1.645\sigma_{\bar{x}}$ will contain the population mean, we say we are 90% confident that the interval $\bar{x} \pm 1.645\sigma_{\bar{x}}$ includes the population mean μ .
- We say that this interval has been established at the 90% confidence level.
- The value 0.90 is referred to as the confidence coefficient.

Confidence Interval of sample mean value for quantitative variable with unknown variance

- If sample size n is **very large** then mean value of a variable with finite variance is an estimation of expectation with **95% Confidence Interval** ($\alpha = 5\%$) given by

$$\left[\bar{X} - 1.96 * \sqrt{S^2 / n}; \bar{X} + 1.96 * \sqrt{S^2 / n} \right]$$

where

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x(i) - \bar{X})^2$$

Interval Estimate of a Population Mean: σ unknown

- If an estimate of the population standard deviation σ cannot be developed prior to sampling, we use the sample standard deviation s to estimate σ .
- This is the σ unknown case.
- In this case, the interval estimate for μ is based on the t distribution.

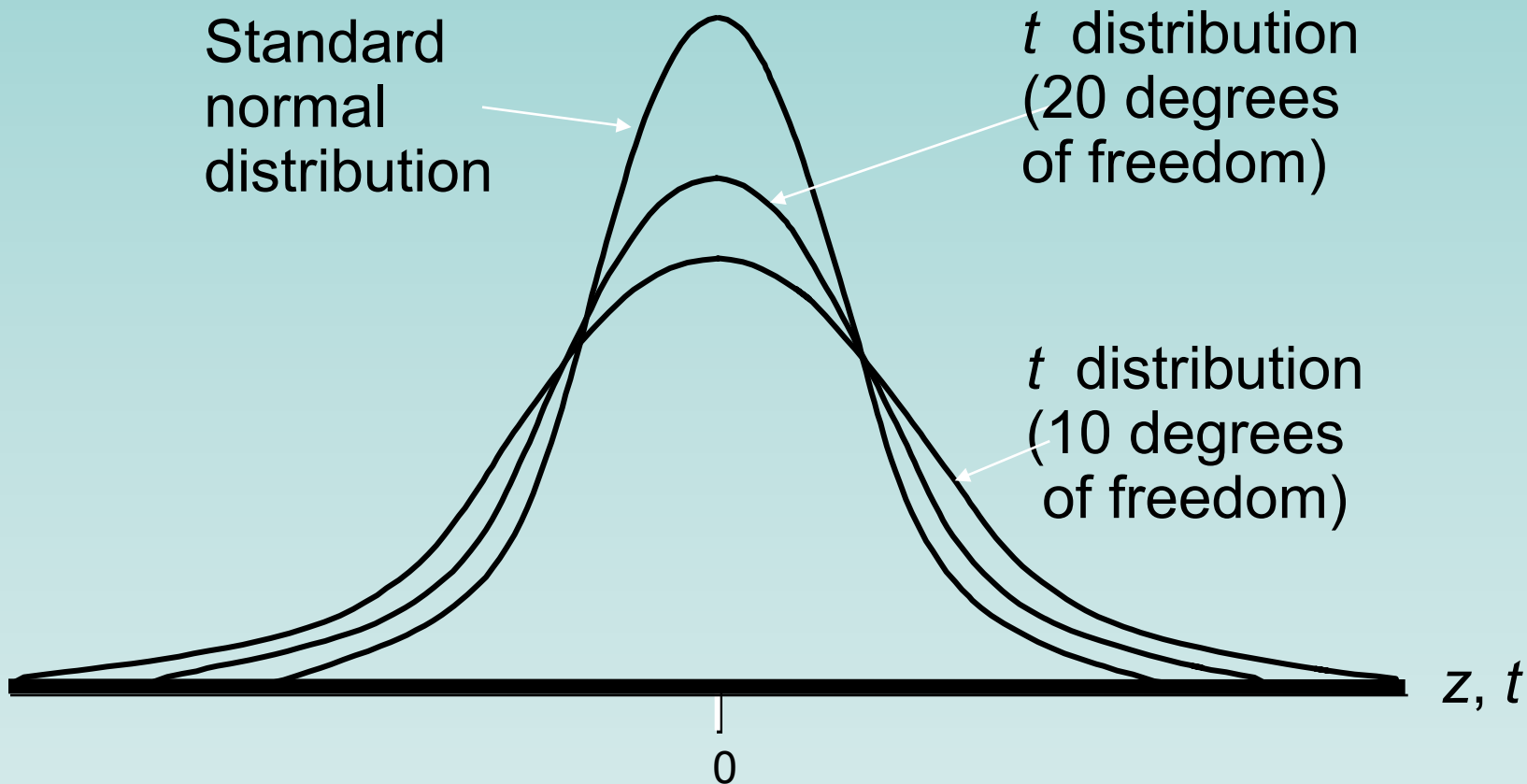
t Distribution

- The t distribution is a family of similar probability distributions.
- A specific t distribution depends on a parameter known as the degrees of freedom.

t Distribution

- A t distribution with more degrees of freedom has less dispersion.
- As the degrees of freedom increases, the difference between the t distribution and the standard normal probability distribution becomes smaller and smaller.

t Distribution



t Distribution

- For more than 100 degrees of freedom, the standard normal z value provides a good approximation to the t value.

Interval Estimate of a Population Mean: σ Unknown

- At 95% confidence, $\alpha = .05$, and $\alpha/2 = .025$.
- $t_{.025}$ is based on $n - 1 = 16 - 1 = 15$ degrees of freedom.

Degrees of Freedom	Area in Upper Tail					
	.20	.10	.05	.025	.01	.005
15	.866	1.341	1.753	2.131	2.602	2.947
16	.865	1.337	1.746	2.120	2.583	2.921
17	.863	1.333	1.740	2.110	2.567	2.898
18	.862	1.330	1.734	2.101	2.520	2.878
19	.861	1.328	1.729	2.093	2.539	2.861
.	.	.	.	.	.	.

t Distribution

Degrees of Freedom	Area in Upper Tail					
	.20	.10	.05	.025	.01	.005
▪	▪	▪	▪	▪	▪	▪
50	.849	1.299	1.676	2.009	2.403	2.678
60	.848	1.296	1.671	2.000	2.390	2.660
80	.846	1.292	1.664	1.990	2.374	2.639
100	.845	1.290	1.660	1.984	2.364	2.626
∞	.842	1.282	1.645	1.960	2.326	2.576

(bottom row is standard normal z values)

Using Excel to Compute t - Distribution

- Excel has two functions for computing cumulative probabilities and x values for any t - distribution:
 - T.DIST is used to compute the cumulative probability given an x value.
 - T.INV is used to compute the x value given a cumulative probability.

- T.DIST(t, DF, cum) = Prob{X<t}
- $X \sim T(DF)$, cum = TRUE

- T.INV = inverse function of T.DIST
- $t_{\alpha/2} = \text{T.INV}(\text{DF}, 1 - \alpha/2)$
- $\text{T.DIST}(t_{\alpha/2}, \text{DF}, \text{TRUE}) = 1 - \alpha/2$

Interval Estimate of a Population Mean: σ Unknown

■ Interval Estimate

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

where: $\bar{x}$ = the sample mean

$1 - \alpha$ = the confidence coefficient

$t_{\alpha/2}$ = the t value providing an area of $\alpha/2$
in the upper tail of a t distribution
with $n - 1$ degrees of freedom

s = the sample standard deviation

n = the sample size

EXCEL PRACTICE

- Sample size:
 - $n = \text{COUNT}(A:A)$
- Mean value:
 - $\text{Mean}(X) = \text{AVERAGE}(A:A)$
- Sample Standard Deviation:
 - $S\text{-SD} = \text{STDEV.S}(A:A)$
 - $= \text{STDEV}(A:A)$
- T - factor
 - $t_{\alpha/2} = \text{T.INV}(n-1, 1- \alpha/2)$
 - $= \text{TINV}(n-1, \alpha)$

Summary of Interval Estimation Procedures for a Population Mean

